

SIPING, China

WMO: 541570

Lat: 43.1203N

Lon: 124.3854E

Elev: 548

StdP: 14.41

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-7.1	-20.9	1.8	-4.7	-17.1	2.2	-2.8	19.4	27.3	16.4	24.1	2.5	200	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	88.2	71.1	86.0	70.7	84.1	70.2	77.0	83.0	75.5	81.5	74.1	80.0	8.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	134.8	80.7	73.7	128.0	79.5	72.2	121.7	78.3	41.0	83.5	39.6	81.8	38.2	80.0	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.5	17.1	14.4	-15.5	91.9	5.7	3.0	-19.6	94.0	-23.0	95.8	-26.2	97.4	-30.4	99.6		
			-15.7	78.6	5.4	2.1	-19.6	80.2	-22.8	81.4	-25.8	82.6	-29.8	84.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.3	8.7	17.2	31.8	48.9	62.1	71.4	75.6	73.0	62.8	47.7	29.2	13.1
	DBStd	24.83	8.79	10.22	9.94	9.12	7.19	5.46	3.90	4.74	6.28	8.81	11.14	9.61
	HDD50	4823	1280	919	567	128	3	0	0	0	2	150	629	1145
	HDD65	8092	1745	1339	1028	487	143	10	0	5	113	538	1074	1610
	CDD50	3093	0	0	5	95	379	643	792	712	385	78	4	0
	CDD65	889	0	0	0	4	54	203	327	253	46	2	0	0
	CDH74	6806	0	0	0	39	492	1641	2542	1764	317	11	0	0
	CDH80	1735	0	0	0	3	111	493	689	404	34	1	0	0
Wind	WSAvg	6.0	4.9	5.8	7.1	8.2	7.6	6.3	5.6	4.8	4.9	5.8	6.0	5.1
Precipitation	PrecAvg	23.7	0.2	0.2	0.6	1.1	2.2	3.7	5.9	5.7	1.7	1.1	0.7	0.4
	PrecMax	34.4	0.8	0.8	1.7	2.6	7.9	8.0	11.6	13.0	5.4	2.8	2.3	1.0
	PrecMin	15.0	0.0	0.0	0.0	0.1	0.3	0.8	1.1	1.3	0.2	0.3	0.0	0.0
	PrecStd	5.2	0.2	0.2	0.4	0.7	1.4	1.7	2.5	3.0	1.1	0.7	0.5	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.6	47.1	63.0	78.9	87.0	91.8	90.8	89.2	83.4	74.8	60.6	43.0
		MCWB	30.6	36.8	47.6	56.0	62.0	67.7	75.3	74.2	69.1	59.9	50.6	38.5
	2%	DB	30.9	41.6	56.0	73.0	83.2	88.2	88.1	86.2	80.4	69.4	55.2	36.4
		MCWB	26.5	33.6	43.2	53.4	60.2	66.2	74.5	73.9	65.5	56.6	47.0	32.1
	5%	DB	26.9	37.8	51.7	69.0	79.7	85.3	85.9	84.2	77.7	65.7	50.4	32.8
		MCWB	23.3	31.2	40.3	52.1	58.8	65.8	73.1	72.7	64.1	54.3	43.4	29.2
	10%	DB	23.5	33.8	47.5	65.1	75.8	82.6	84.0	82.1	75.1	62.3	46.1	28.9
		MCWB	20.3	28.1	37.8	49.2	57.1	65.5	72.8	71.7	62.5	52.7	40.7	25.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.5	38.3	49.1	60.3	66.3	73.3	79.4	78.8	72.7	62.1	51.7	38.9
		MCDB	34.2	44.9	60.6	70.8	81.2	83.8	87.0	84.8	79.9	70.3	58.2	42.6
	2%	WB	26.8	34.4	44.8	56.0	63.5	71.0	77.4	76.9	69.6	59.1	47.6	32.9
		MCDB	30.4	40.4	53.8	69.6	76.9	80.8	83.8	82.6	75.9	66.8	54.0	35.7
	5%	WB	23.4	31.5	41.3	52.8	61.2	69.7	76.1	75.4	67.2	56.3	44.4	29.4
		MCDB	26.7	37.5	50.7	66.6	73.9	79.1	82.4	80.9	74.2	64.0	49.7	32.7
	10%	WB	20.1	28.3	38.2	50.3	59.3	68.2	74.8	74.1	65.0	53.9	40.9	25.4
		MCDB	23.5	33.8	47.0	62.6	72.1	77.4	81.0	79.3	72.2	61.0	45.7	28.8
Mean Daily Temperature Range	5% DB	MDBR	17.1	17.6	17.3	19.7	19.4	16.7	13.8	15.0	20.2	19.6	15.8	16.0
		MCDBR	20.7	19.3	21.6	24.2	23.6	21.3	16.8	17.5	22.6	22.9	18.9	18.6
		MCWBR	17.7	13.9	13.2	12.1	9.8	7.6	6.8	7.5	10.2	12.3	12.9	16.4
	5% WB	MCDBR	19.9	18.3	20.2	20.1	18.9	15.9	13.2	13.0	17.3	19.3	16.7	18.6
		MCWBR	17.4	14.1	13.1	11.7	9.6	6.9	6.9	7.1	9.6	12.1	12.8	17.0
Clear-Sky Solar Irradiance	taub	0.338	0.383	0.455	0.553	0.620	0.689	0.660	0.577	0.497	0.512	0.418	0.355	
	taud	2.064	1.971	1.862	1.707	1.625	1.575	1.670	1.832	1.946	1.806	1.958	2.031	
	Ebn at Noon	239	247	243	230	218	204	208	221	229	199	204	217	
	Edh at Noon	34	44	57	72	80	84	76	62	52	52	38	32	
All-Sky Solar Radiation	RadAvg	728	1031	1379	1629	1818	1716	1591	1532	1407	1021	696	594	
	RadStd	53	71	67	111	138	156	141	121	92	69	64	46	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page